

Future trends in (big) data management and Integration

reference:

Principles of Data Integration
by A. Doan, A. Halevy, and Z. Ives
Morgan Kaufmann

Uncertain Data

- Databases are assumed to represent *certain data*: a tuple in the database is true, any tuple NOT in the database is false (Closed World Assumption)
- Real life is not as certain! Examples:
 - Observations (e.g. sensor readings) may be *unreliable*
 - A person might be still *undecided* as to a certain choice
- *Uncertain databases* of various kinds (e.g. based on probability) attempt to model uncertain data and to answer queries in an uncertain world

Uncertainty in Data Integration

- Data itself may be uncertain (e.g. extracted from an unreliable source)
- Mappings might be approximate (e.g. created by relying on automatic ontology matching)
- Reconciliation is approximate
- Approximate mediated schema
- Imprecise queries, such as keyword-search queries, are approximate

Uncertain Databases

- Whatever the semantics of uncertainty (e.g. probabilistic or fuzzy...) an uncertain database describes a set of possible worlds, i.e. all those worlds that are compatible with the probability or the fuzziness value
- What is uncertain? A **value**? A **tuple**? A **table**?
- Example: Assign each tuple a probability. Then the probability of a possible world is the product of the probabilities of the tuples.
- However, this still does not represent possible correlations between tuples.

The steps of Data Integration ✓

Schema Reconciliation

This problem is certainly not only for structured data !

Schema reconciliation: mapping the **data structure**

Record Linkage

Record linkage: data matching based on **the same content**

Data Fusion

Data fusion: reconciliation of **non-identical content**

Entity Resolution: recall

- **Entity resolution** (ER) is the problem of accurately identifying multiple, differing, and possibly contradictory representations of unique real-world entities in data.
- This problem is studied under different names, viz.: entity resolution (ER), merge/purge, deduplication, **record linkage**, etc.
- The causes of multiple representations of the same real-world entity are:
 - **Noise**, i.e. 'small errors' (e.g., a typo 'Peter' for 'Peters');
 - **Inconsistencies**, i.e. conflicting and/or different values for properties/relations of an object.

Researchers are still looking for efficient solutions for detecting duplicates in data

- A common approach: use of record-matching rules (rule-based Entity Resolution), i.e., constraints that state:
“if any two records are similar on properties $P_1, \dots, P_n$, then they refer to the same entity”.
- Problems:
 - it is difficult to generate record-matching rules;
 - it is challenging to define the bounds of similarity for approximate-match conditions;
 - efficient support for approximate-match conditions is non-trivial.
 - one-to-one record matching too expensive (quadratic) →
example of a solution: BLOCKING, i.e., divide the two datasets to compare into blocks of tuples in such a way that records within each group hold a high chance of being linked in the subsequent linkage process

Data Provenance

- Also called data lineage or data pedigree.
- Knowing where the data come from, and how they were produced, is very important in many cases and is useful anyway. E.g., an information extractor might be unreliable, or one data source is more authoritative than others.
- Sometimes it can also be difficult, depending on the metainformation that has been collected during the life of those data.

Data Provenance

- Provenance of a data item records “where it comes from”:
 - Who created it
 - When it was created
 - How it was created - as a value in a database, as the result of a computation, coming from a sensor, etc...
- The database community models provenance in terms of **how the datum was derived from the original source databases**, the rest is left to the application (it is assumed to be domain-dependent)

Two Viewpoints on Provenance

- *Provenance as Annotations (meta-information) on Data:* models provenance as a set of annotations describing how each data item was produced. These annotations can be associated with tuples or values
- *Provenance as a Graph of Data Relationships:* it models provenance as a (hyper)graph, with tuples as vertices. Each possible direct derivation of a tuple from a set of source tuples is a hyper-edge connecting the source and derived tuples

The two views are equivalent, and it is possible to convert one into the other as convenient

Uses of provenance information

- Explanations
- Scoring of sources and data quality
- Influence of sources on one another

Uncertainty, Provenance, and Cleaning

- Exploit data usage, user feedback and data quality info to assess uncertainty and automate cleaning
- Develop formalisms, data models and query primitives to capture the semantics of uncertainty propagation

Using Crowdsourcing

- Some checks are very simple for humans but hard for a computer, e.g.: image contents, Web content extraction
- **Crowdsourcing** can provide powerful solutions to traditionally hard data integration problems (e.g. wrapping, check correctness of schema mappings, record matching, etc.)
- **Amazon Mechanical Turk (MTurk)** is a crowdsourcing marketplace that connects businesses (requesters) with a distributed workforce (workers) to complete tasks that computers can't do. These are often small, repetitive tasks, or "microtasks," that can be completed over the internet for payment, such as data categorization, image annotation, or survey participation. The platform allows companies to outsource processes and harness the collective intelligence of a global workforce on-demand.

Building Large-Scale Structured Web Databases

Examples:

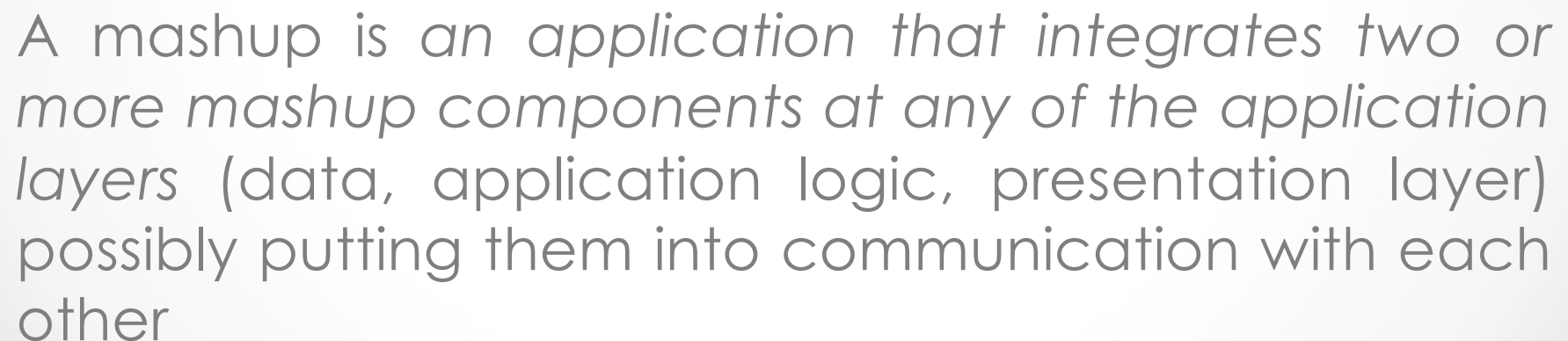
- Google Scholar integrates bibliographic sources to build a citation database for millions of publications.
- Physicists want to integrate the results of their experiments
- Doctors and biologists want to integrate sources such as PubMed, Grants, and Clinical Trials to build biomedical databases.
- Companies integrate product information for searching, analysis and marketing purposes.

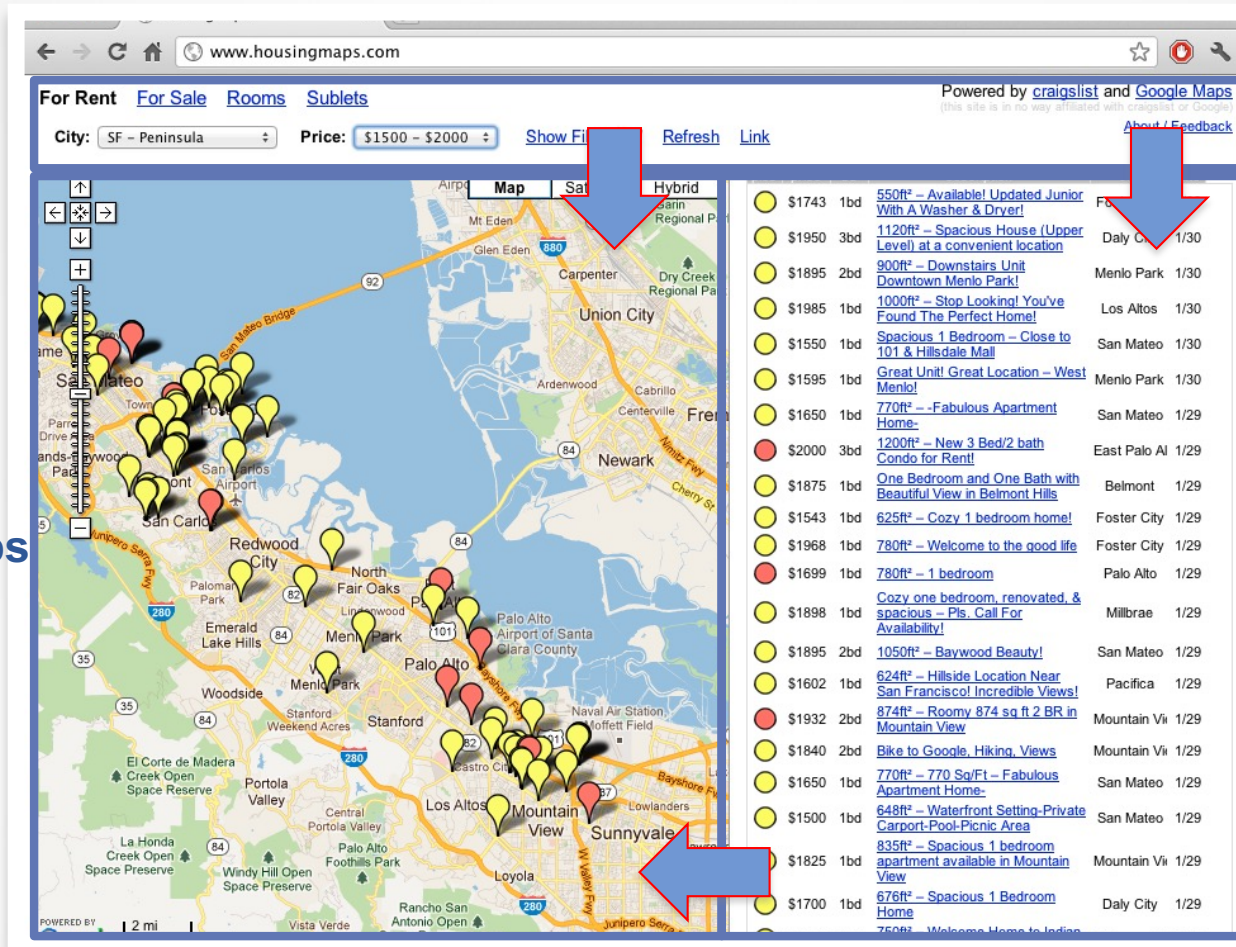
Lightweight Integration

Many data integration tasks are transient:

- We may need to integrate data from multiple sources **to answer a question asked once or twice**. The integration needs to be done **quickly** and **by people without technical expertise** (e.g. a disaster response situation in which reports are coming from multiple data sources in the field, and the goal is to corroborate them and quickly share them with the affected public)
- **Problems** typical of lightweight data integration:
 - locating relevant data sources
 - assessing source quality
 - helping the user understand the semantics
 - supporting the process of integration.
- **Machine learning and other techniques** can be used to amplify the effects of human input, through semi-supervised learning, where small amounts of human data classification, plus large amounts of additional raw (“unlabeled”) data, are used to train the system.

Cinzia Cappiello, Florian Daniel, Maristella Matera)





Craigslist

Provides for the synchronized exploration of housing offers from *craigslist.com* and maps by *Google Maps*

Integration is the added value provided by the mashup

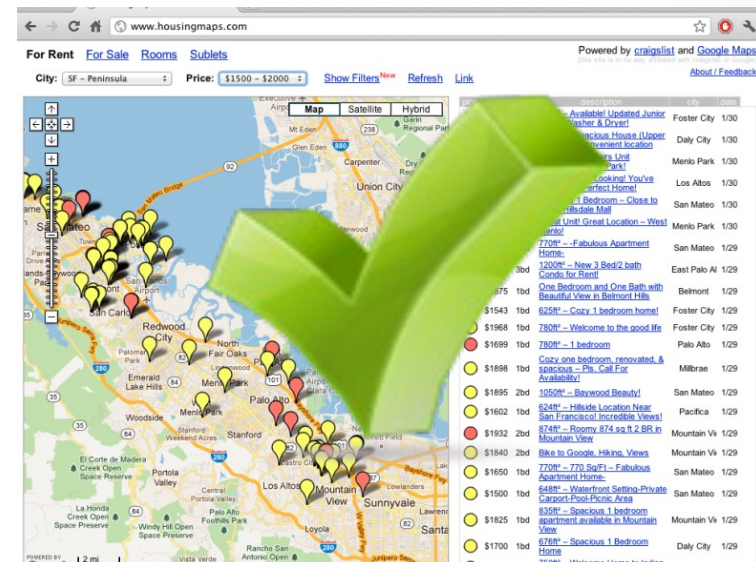
Key elements

Mashup component is any piece of data, application logic and/or user interface that can be reused and that is accessible either locally or remotely (e.g., Craigslist and Gmaps)

Mashup logic is the internal logic of operation of a mashup; it specifies the invocation of components, the control flow, the data flow, the data transformations, and the UI of the mashup



No added value



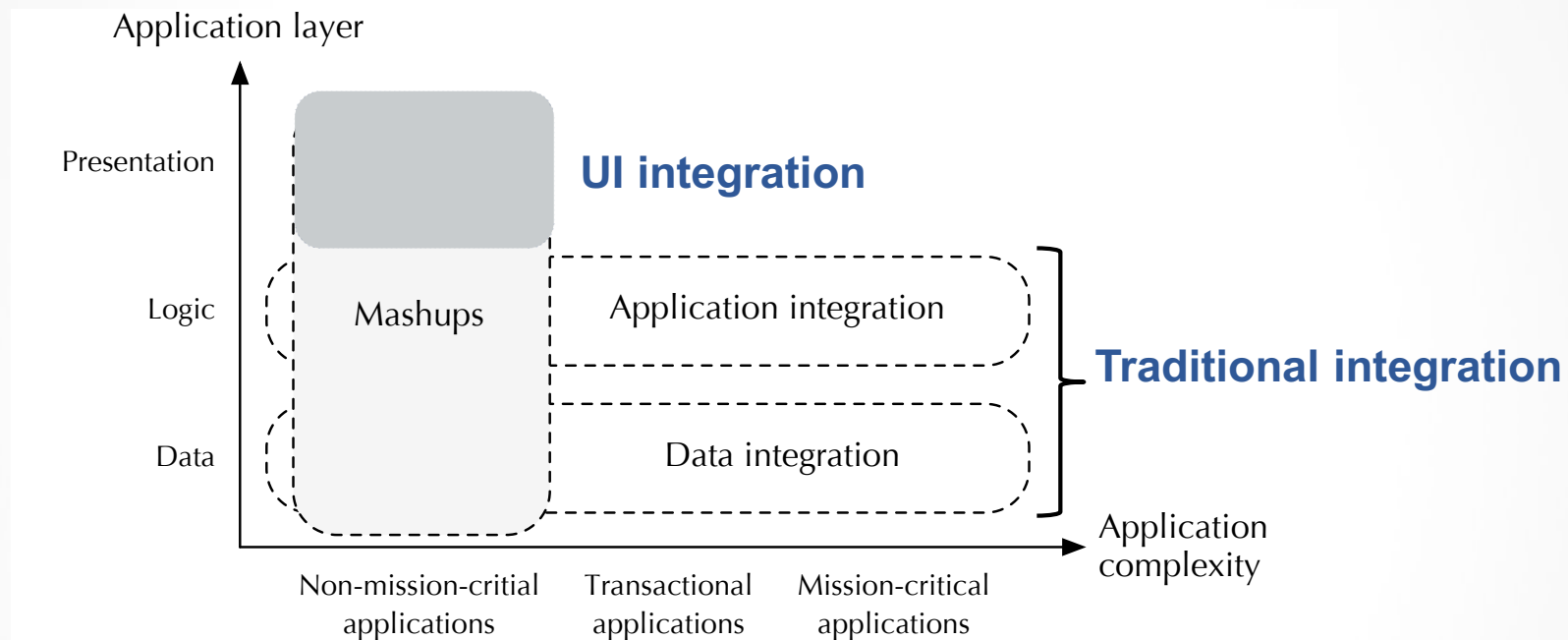
Additional information, functions, visualizations!

Types of mashup

- Data mashups
 - Fetch data from different resources, process them, and return an integrated result set
- Logic mashups
 - Integrate functionality published by logic or data components
- User Interface (UI) mashups
 - Combine the component's native UIs into an integrated UI; the components' UIs are possibly synchronized with each other
- Hybrid mashups
 - Span multiple layers of the application stack, bringing together different types of components inside one and a same application;
 - integration happens at more than one layer

Data mashup vs. data integration...

- Data mashups are a Web-based, lightweight form of data integration, meant to solve different problems
- Covering the “long tail” of data integration requirements
 - Very specific reports or ad-hoc data analyses
 - Simple, ad-hoc data integrations providing “situational data” that meet short term needs
 - Non-mission-critical integration requests



Mashup positioning in relation to other integration practices

Mashups introduce integration at the presentation layer and typically focus on non-mission-critical applications

Pay-as-You-Go Data Management

- Data integration on the Web may be an extreme example motivating **pay-as-you-go** data management.
- In contrast to the other kinds of data integration systems, pay-as-you-go systems try to avoid the need for the initial setup phase that includes creating the mediated schema and source descriptions.
- The goal is to offer a system that provides useful services on a collection of heterogeneous data with very little up-front effort.

Climbing on the giants' shoulders 😊

- Many people are trying to access the same data sources and perform similar tasks. Examples:
 - when someone extracts a structured data set (excel or DB) from semistructured data it means that data is “easily structurable”, and we might also copy the names of the columns
 - When someone manually combines two data sets, we can infer that the two sources are related to each other and think about making a join between “similar” columns.
- Cloud-based tools for data management might provide such services since they log the activities of many users.
- It would be useful to record the actions of tons of users butproblems with privacy!

Visualizing Integrated Data

- Visualize important patterns in the data instead of an infinite number of rows
- Immediately see discrepancies between the data in the sources.
- During the integration process, show a subset of the data that was not reconciled correctly
- When browsing different collections of data to be integrated, visually show the search results and evaluate their relevance to the specific integration task
- Visualization of the data provenance.

Integrating Social Media

- Social media data are often noisy (spam and low-quality data)
- Transient nature of such data and users: identifying quality data and influential users is difficult
- The data lack context and therefore it is hard to interpret
- Data often arrive as high-speed streams that require very fast processing

Cluster- and Cloud-Based Parallel Processing and Caching

- Most query engines, schema matchers, storage systems, query optimizers, have been developed for operation on a single server or a few servers
- Most algorithms (schema matching, entity resolution, data cleaning, indexing, etc.) are based on assumptions of limited scale (sometimes main memory):
 - need to be tackled in a much more parallelizable and scalable way
 - need to be redesigned to exploit the power of large clusters

Conclusion

- Data management and Integration: a fascinating and ever-challenging problem, lending itself to a multitude of solutions
- A still challenging problem, in terms of infrastructure, integration, data cataloguing, data movement and privacy.
- Making sense of Big Data: still the BIG problem

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Ethics in Data Science

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The promise of Big Data: what about ethics?

- 5Vs: volume, velocity, variety, veracity, value
- unprecedented data collection capabilities
- enormous computational power
- massively parallel processing
- improve people's lives, e.g., recommendation
- accelerate scientific discovery, e.g., medicine
- boost innovation, e.g., autonomous cars
- transform society, e.g., open government

Ethics in Data Management

- As data have an impact on almost every aspect of our lives, it is more and more important to understand the nature of this effect
- With search and recommendation engines, the web can influence our lives to a great extent, e.g. by recommending interesting jobs only to white males, discriminating as an effect of biased data or algorithms
- With statistics used everywhere, it may happen that very critical decisions be taken without taking their ethical consequences into account

BUT....

THE WALL STREET JOURNAL.

WHAT THEY KNOW

Websites Vary Prices, Deals Based on Users' Information

By JENNIFER VALENTINO-DEVRIES,
JEREMY SINGER-VINE and ASHKAN SOLTANI
December 24, 2012

It was the same Swingline stapler, on the same Staples.com website. But for Kim Wamble, the price was \$15.79, while the price on Trude Frizzell's screen, just a few miles away, was \$14.29.

A key difference: where Staples seemed to think they were located.

lower prices offered to buyers who live in more affluent neighborhoods

<https://www.wsj.com/articles/SB10001424127887323777204578189391813881534>

WHAT PRICE WOULD YOU SEE?



BUT....

theguardian

Samuel Gibbs

Wednesday 8 July 2015 11.29 BST

Women less likely to be shown ads for high-paid jobs on Google, study shows

Automated testing and analysis of company's advertising system reveals male job seekers are shown far more adverts for high-paying executive jobs



One experiment showed that Google displayed adverts for a career coaching service for executive jobs 1,852 times to the male group and only 318 times to the female group. Photograph: Alamy

The AdFisher tool simulated job seekers that did not differ in browsing behavior, preferences or demographic characteristics, except in gender.

One experiment showed that Google displayed ads for a career coaching service for "\$200k+" executive jobs **1,852 times to the male group and only 318 times to the female group**. Another experiment, in July 2014, showed a similar trend but was not statistically significant.

<https://www.theguardian.com/technology/2015/jul/08/women-less-likely-ads-high-paid-jobs-google-study>

BUT....

THE WALL STREET JOURNAL.

Are Workplace Personality Tests Fair?

Growing Use of Tests Sparks Scrutiny Amid Questions of Effectiveness and Workplace Discrimination

By **LAUREN WEBER** and **ELIZABETH DWOSKIN**

Sept. 29, 2014 10:30 p.m. ET



Cyle Behm accused Kroger and six other companies of discrimination against the mentally ill through their use of personality tests. TROY STAINS FOR THE WALL STREET JOURNAL

The Equal Employment Opportunity commission is **investigating whether personality tests discriminate against people with disabilities.**

As part of the investigation, officials are trying to determine if the tests **shut out people suffering from mental illnesses** such as depression or bipolar disorder, even if they have the right skills for the job.

<http://www.wsj.com/articles/are-workplace-personality-tests-fair-1412044257>

BUT....



A commercial tool COMPAS automatically predicts some categories of future crime to assist in bail and sentencing decisions. It is used in courts in the US.



The tool correctly predicts recidivism **61% of the time.**

Blacks are almost twice as likely as whites to be labeled a higher risk but not actually re-offend.

The tool makes **the opposite mistake among whites**: They are much more likely than blacks to be labeled lower risk but go on to commit other crimes.

<https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>

The illusion....

Big Data processing is based on algorithms, thus it must be objective!

Unfortunately:

- Algorithms are based on data, and data may contain bias
- Often the algorithms produce opaque processes (e.g. deep neural networks) that *cannot be explained* → AI research area: *EXPLAINABILITY*

→ Technology must be accompanied by *ethical and legal considerations and competence*

Ethics in Data Management:

RESPONSIBILITY

It is up to the data scientists to

- identify which datasets can genuinely help answering some given question
- understand their contents
- choose the most appropriate *knowledge extraction technique* (search, query, or data analysis/AI methods) to obtain a fair result

This sequence of choices may strongly influence the process, *and biased results might be obtained.*

Ethics in Data Management

- What role should the database and AI communities, and computer engineers in general, play ?
- Should *ethical dimensions* as *data protection, fairness, diversity, transparency* become fundamental issues that can drive new ideas, and should they be enforced in all processes?

**Data
Protection**



Transparency



Diversity



Fairness



Data protection

- Traditional knowledge extraction systems, including database systems, search engines, data warehouses, etc., hardly pay specific attention to ethically sensitive aspects of knowledge extraction processes and their outcomes.
- Such aspects are now becoming prominent, especially with regard to the protection of human rights and their consequences in normative ethics. These demands are broadly reflected into codes of ethics for companies and computer professionals, and also in legally binding regulations such as the EU General Data Protection Regulation (GDPR):

<https://www.eugdpr.org/>

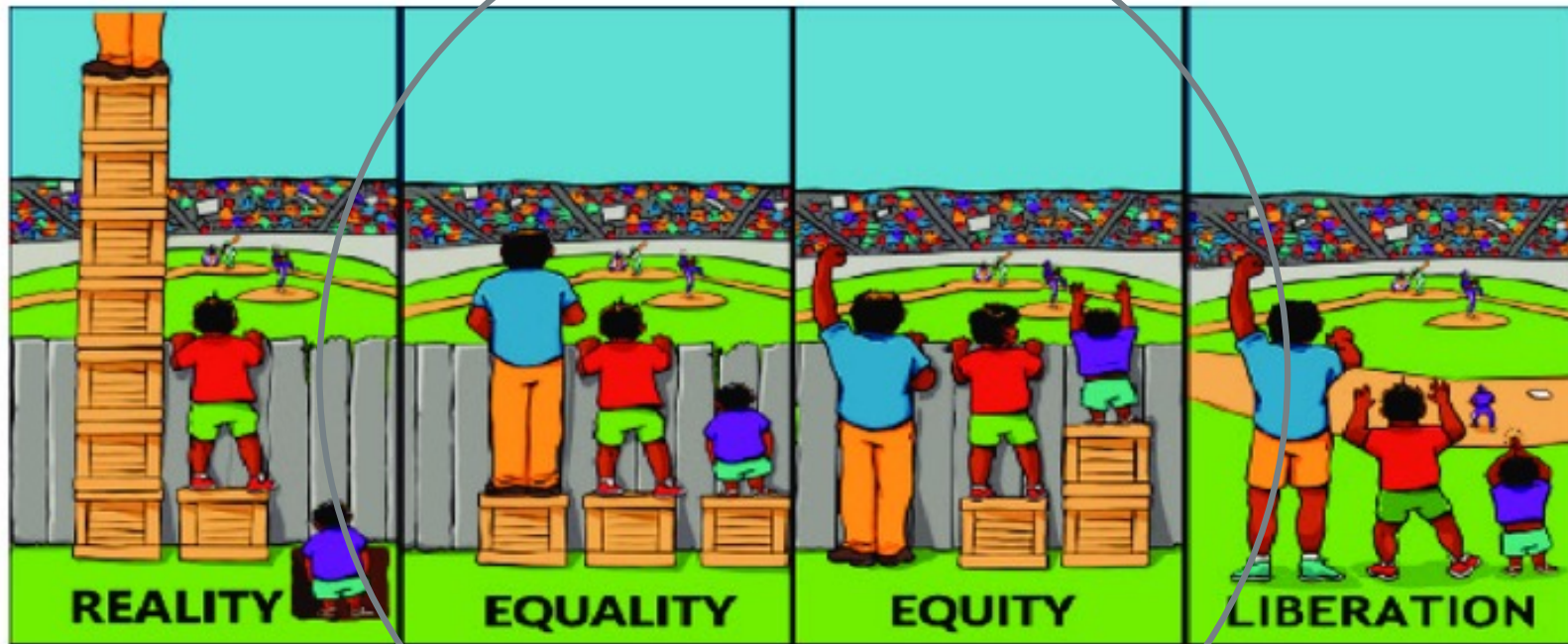
- GDPR unifies data protection laws across all European Union members, defining a comprehensive set of rights for EU citizens, describing the requirements for companies and organizations for collecting, storing, processing and managing personal data.
- Following a 2-year post-adoption grace period, the GDPR has become fully enforceable throughout the European Union in May 2018.

Fairness

Fairness of data is defined as **the lack of bias**

- Recently its importance has been acknowledged e.g. due to the unsettling consequences of training systems with biased data (e.g. in the examples seen above)
- Various “categories” of fairness have been defined, e.g. equality vs. equity

Many interpretations of the notion of fairness



The difference between the terms equality, equity, and liberation, illustrated; © Interaction Institute for Social Change | Artist: Angus Maguire

(H. Jagadish)

Transparency

Transparency is the ability to interpret the information extraction process in order to verify which aspects of the data determine its results. In this context, transparency metrics can use the notions of

- *data provenance*, describing where the original data come from;
- *explanation*, describing how a result has been obtained.
- Transparency may conflict with Data Protection
-

Diversity

Diversity is the degree to which *different kinds of objects* are represented in a dataset.

Several

metrics of diversity are proposed.

- Ensuring diversity at the beginning of the information extraction process may be useful for enforcing fairness at the end.
- The diversity dimension may conflict with data quality needs, that prioritize the use of few, high-reputation sources.

Ethics and Data Quality

- Data quality is a typical ethical requirement: we *could never trust a piece of information* if it did not have the typical data quality properties.
- Yet, we can also assert the opposite: that *data should conform to a high ethical standard*, for it to be considered of good quality.
- Hence, the satisfaction of the ethical requirements is actually necessary to assert the quality of a result
- It is the responsibility of the system designer, and of the person/company that ordered the job, to ensure that the (necessary) ethical properties are satisfied.

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